

Externality

A cost or benefit borne by a third party who is not party to the transaction

GLOSSARY 05 - SHEET 5 / 8

PROFESSIONAL DEFINITION

An externality is a cost or benefit arising from an economic activity that falls on a party who neither chose to incur it nor is compensated for it, and which therefore does not appear in the price of the activity. Because the decision-maker does not bear the full cost, activity generating negative externalities is over-supplied relative to the socially efficient level; correction requires that the external cost be internalised through regulation, taxation or assignment of rights (Pigou, 1920; Coase, 1960).

WHO DEFINED IT ORIGINALLY

Arthur C. Pigou formalised the concept and the corrective tax in *The Economics of Welfare* (1920) at the University of Cambridge. Ronald H. Coase (1960) established that with well-defined property rights and low transaction costs the parties may bargain to an efficient outcome without a tax, for which he received the 1991 Nobel Memorial Prize.

WHY THIS DEFINITION WAS DEVELOPED

Classical price theory assumed that prices carried all relevant costs, which was contradicted by industrial pollution. Pigou needed a mechanism reconciling private optimisation with social efficiency. Coase's later contribution reframed the question from who is at fault to which allocation of rights minimises total cost, which changed the design of environmental and technology regulation.

FIGURE 1 - PRIVATE COST AGAINST SOCIAL COST



Figure 1. The gap between the two is the externality. Because the deploying firm optimises against private cost, activity generating negative externalities is over-supplied until a rule, a tax or an assignment of rights internalises the difference (after Pigou, 1920; Coase, 1960).

HOW IT IS USED

Identify who bears the costs of an AI deployment other than the deploying firm and its customers - energy and water consumption, labour transition costs, effects on those subject to automated decisions - and state them in the appraisal even where they are not priced.

WHEN IT IS USED

In AI impact assessment, in environmental appraisal of computing infrastructure, in regulatory analysis, and in any business case whose benefits accrue internally while some costs do not.

BY WHOM IT IS USED

Policy analysts and regulators, sustainability and environmental social and governance functions, boards assessing societal impact, and economists appraising public investment.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor position had no separate category for uncompensated third-party effects, which meant they entered analysis only as moral objection. Pigou made the effect a quantity to be measured and priced; Coase then showed that the tax is one of several instruments and not always the efficient one. For AI the framework is directly applicable: energy consumption, labour displacement and error costs borne by decision subjects are all external to the deployment decision unless a rule makes them internal.

RELATED DEFINITIONS IN THIS COURSE

[Green AI - Unit 5](#)
[Labour Market Displacement - Unit 5](#)
[Life Cycle Assessment - Unit 5](#)
[Benefits Realisation - Unit 4](#)
[AI Ethics - Unit 6](#)

SOURCES (HARVARD REFERENCING STYLE)

Coase, R.H. (1960) 'The problem of social cost', *Journal of Law and Economics*, 3, pp. 1-44.

Pigou, A.C. (1920) *The Economics of Welfare*. London: Macmillan.

Stern, N. (2007) *The Economics of Climate Change: The Stern Review*. Cambridge: Cambridge University Press.